



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination 2018
Higher 2

CANDIDATE
NAME

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CIVICS
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H2 BIOLOGY

9744/01

Paper 1 Multiple Choice

20 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid / tape.

Write your name, civics group and index number on the Multiple Choice Answer Sheet provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

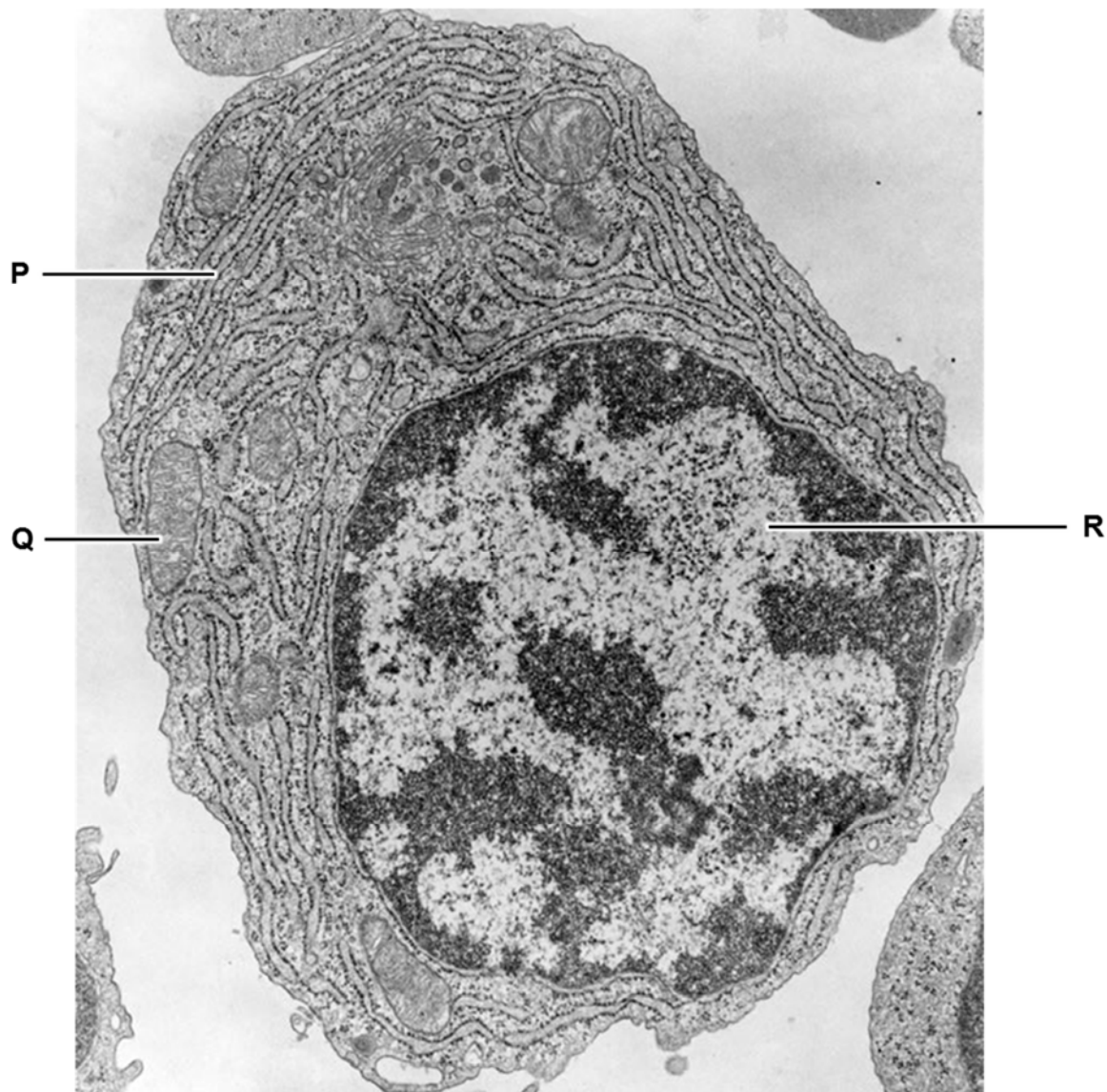
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

At the end of the examination, you may bring home the Paper 1 question paper.

QUESTION 1

The diagram shows an electron micrograph of a cell.



Which row correctly describes structures **P – R**?

	P	Q	R
A	synthesis of ribosomes	substrate level phosphorylation	active replication of genes
B	provision of large surface area for attachment of ribosomes	formation of ATP from light energy	active transcription of genes
C	synthesis of membrane proteins	oxidative decarboxylation	active transcription of genes
D	transport of proteins to Golgi apparatus	modification of mRNA transcripts	active replication of genes

QUESTION 2

A ribosome consists of a large and a small subunit, each subunit containing ribosomal RNA (rRNA) complexed with proteins.

Which sequence of events concerning ribosomes is correct?

- A** Within the nucleolus, rRNA and proteins are synthesised and subunits are formed. They become membrane bound as they are exported through the nuclear envelope to the cytoplasm and rough endoplasmic reticulum (rER).
- B** rRNA and proteins are synthesised in the Golgi body and are transported to the nucleolus for subunit formation.
- C** rRNA is synthesised in the nucleolus and proteins are synthesised by the rough endoplasmic reticulum (rER). Subunit formation occurs within the cytoplasm for free ribosomes and on the surface of the rER for attached ribosomes.
- D** rRNA synthesised within the nucleolus is complexed with proteins that have been imported from the cytoplasm. The subunits formed are exported to the cytoplasm via the nuclear pores.

QUESTION 3

The following are all observations about cell surface membranes.

- 1 Phospholipids labelled with radioactive phosphate groups change position over time.
- 2 Proteins labelled with fluorescent dyes change position over time.
- 3 Lectins are proteins that bind to polysaccharides. They only attach to the outside of membrane samples.
- 4 Electron microscope images of membranes fractured by freezing show a large number of regularly arranged particles interrupted by larger particles.
- 5 Some membranes contain more cholesterol and unsaturated fatty acid chains than other membranes.
- 6 Some proteins can only be separated from the membrane by disrupting the membrane with detergent.

Which observations provide evidence for the fluid mosaic model of cell surface membranes?

- A** 1, 2 and 4 **B** 1, 2 and 6 **C** 3, 4 and 5 **D** 3, 5 and 6

QUESTION 4

Which row gives the correct description of **both** a collagen molecule and a collagen fibre?

	collagen molecule	collagen fibre
A	α and β polypeptide chains forming a double helix, held together by peptide bonds	molecules of collagen arranged randomly to each other, linked by hydrogen bonds
B	a polypeptide chain, with repeating amino acids, forming an α -helix	three molecules of collagen, forming a triple helix, held together by hydrogen bonds
C	a polypeptide chain, with repeating amino acids, forming a helix	three α helical collagen molecules, forming a triple helix, held together by covalent bonds
D	three helical polypeptide chains, forming a triple helix, held together by hydrogen bonds	molecules of collagen lying parallel and cross-linked to each other

QUESTION 5

Glycogen is a compact molecule with many branches.

Which types of glycosidic bond result in the biologically important characteristics of the glycogen molecule?

	characteristic of glycogen molecule				
	branching chains	coiled chains	compact shape	large osmotically neutral molecule	rapid hydrolysis
A	1,4	1,4	1,4	1,6 and 1,4	1,6
B	1,6	1,4	1,6 and 1,4	1,6 and 1,4	1,6
C	1,6	1,6	1,6 and 1,4	1,4	1,4
D	1,6 and 1,4	1,6 and 1,4	1,6	1,4	1,6 and 1,4

QUESTION 6

Two polynucleotide strands make up a DNA molecule.

What is a correct description?

- A** The percentage of cytosine is 50% of that of guanine in the whole molecule.
- B** The percentage of cytosine is the same as that of guanine in each strand.
- C** The percentage of cytosine is the same as that of guanine in the whole molecule.
- D** The percentage of cytosine is the same in each strand of the molecule.

QUESTION 7

The table contains results recorded by a student from an investigation into the effect of temperature on an enzyme-catalysed reaction. All other variables were standardised.

temperature / °C	rate of reaction / arbitrary units
10	3
20	7
30	16
40	33
50	32
60	14

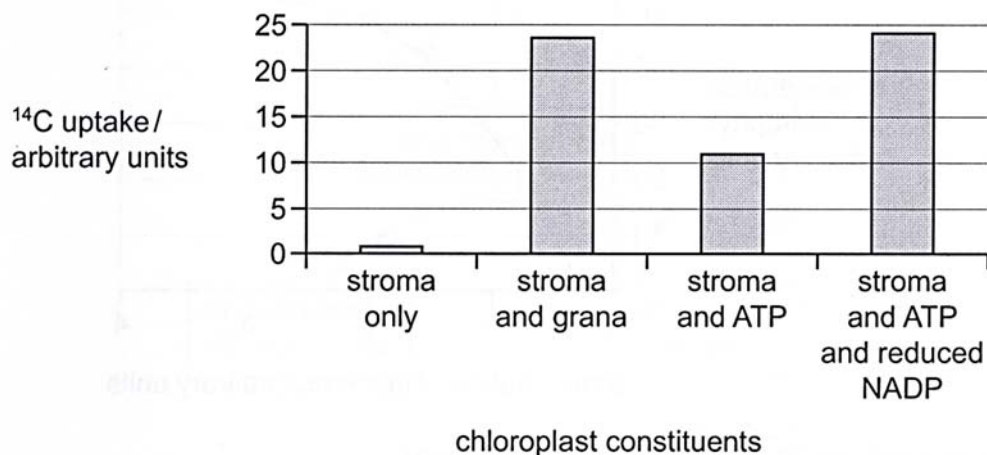
What can be best concluded from the results?

- A The enzyme had the highest kinetic energy at 40°C.
- B The data for 50°C was anomalous.
- C The optimum temperature was between 30°C and 50°C.
- D The enzyme was held by only disulfide bonds at 60°C.

QUESTION 8

Carbon dioxide labelled with radioactive carbon, ^{14}C , was used to study the processes involved in carbon fixation. The uptake of ^{14}C by stroma, alone or mixed with other constituents of chloroplasts, was measured after a 60-second exposure to $^{14}\text{CO}_2$.

The results are shown below.



Which conclusion can be reached from these results?

- A ATP is half as efficient as reduced NADP at uptake of $^{14}\text{CO}_2$.
- B $^{14}\text{CO}_2$ is taken up easily by grana.
- C Stroma can easily take up $^{14}\text{CO}_2$ when ATP and reduced NADP are present.
- D Stroma cannot fix carbon into a 3C compound.

QUESTION 9

Some apples can be stored in controlled atmospheric conditions for up to a year. Taste and texture are maintained by using conditions that reduce the production of a fruit-ripening plant hormone while limiting the build-up of ethanol. Ethanol damages the fruit.

The storage conditions needed include low temperature (1°C), high carbon dioxide concentration (1.2%) and low oxygen concentration (0.9%).

Why are these conditions needed?

- 1 Low oxygen concentration favours anaerobic respiration.
- 2 Enzyme activity is reduced.
- 3 Conversion of sugar to ethanol is minimised.
- 4 High carbon dioxide concentration promotes photosynthesis.

A 1, 2 and 3 **B** 1, 2 and 4 **C** 2 and 3 only **D** 3 and 4 only

QUESTION 10

In plants, dinitroaniline herbicides are thought to prevent the formation of microtubules from the protein, tubulin. These microtubules form the spindle of cells undergoing mitosis.

The herbicides appear to have no effect on mammalian cells, but do affect parasitic protocists that cause diseases in humans, such as malaria. Protocists are eukaryotes.

Which statement is consistent with this information?

- A** Dinitroaniline herbicides are most likely to act in the G1 and G2 phases of the cell cycle.
- B** Dinitroaniline herbicides prevent plant cells from entering the M phase of the cell cycle.
- C** Plant cells and protocists have similar enzymes controlling the formation of microtubules from the protein, tubulin.
- D** There are more similarities in spindle formation in mammals and protocists than in protocists and plants.

QUESTION 11

Which statement helps to explain how genetically identical cells are produced during the mitotic cell cycle?

- A** There is only one origin of replication, which ensures that DNA replication begins at a controlled site.
- B** Checkpoints of cell division are carefully regulated.
- C** Complementary base pairing occurs during DNA replication.
- D** RNA polymerase can correct some mistakes during transcription.

QUESTION 12

Which of the following statements explain the difference in the direction in which the two strands of a DNA molecule are synthesised?

- 1 The replication of DNA is semi-conservative.
- 2 DNA polymerase can only add deoxyribonucleotides to the 3'OH group.
- 3 Each DNA molecule consists of two anti-parallel polynucleotides.
- 4 The synthesis of the lagging strand requires many more RNA primers.

A 1 and 2 only **B** 2 and 3 only **C** 1, 2 and 4 only **D** 2, 3 and 4 only

QUESTION 13

Which statement(s) describe(s) how a gene mutation can lead to the production of a non-functional protein?

- 1 During transcription an incorrect nucleotide is added to a DNA molecule.
- 2 A codon in the mRNA transcribed from the mutated gene is changed.
- 3 The order of the bases in an anticodon on tRNA is altered during translation.
- 4 The sequence of nucleotides in the promoter of the gene is altered

A 2 only **B** 1 and 2 only **C** 2 and 4 only **D** 2, 3 and 4 only

QUESTION 14

Exceptions to the universal genetic code are found in mammalian mitochondria, as shown in the table.

mRNA codon	in mammalian cytoplasm, codes for	in mammalian mitochondria, codes for
AGA	arginine	stop
AGG	arginine	stop
AUA	isoleucine	methionine
UGA	stop	tryptophan

A short length of DNA triplet code with the following base sequence was transcribed.

TATTCTTCCACT

How many peptide bonds would be formed by ribosomes during translation in mammalian cell cytoplasm and in mammalian mitochondria?

	mammalian cell cytoplasm	mammalian mitochondria
A	2	1
B	2	0
C	3	0
D	3	1

QUESTION 15

The following discoveries were made about myostatin, a protein that is produced in mammalian skeletal muscle cells.

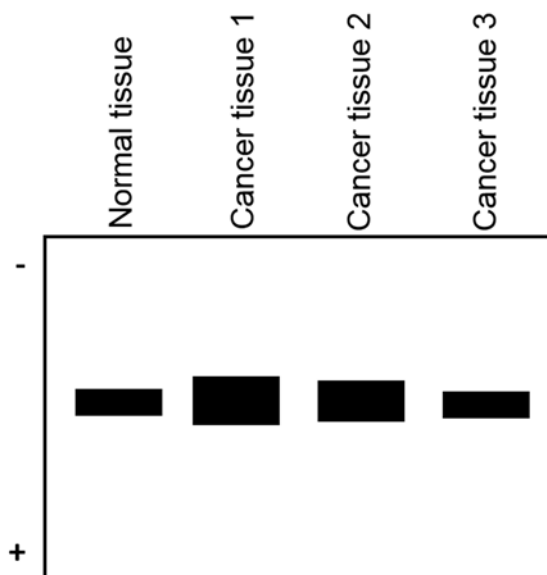
- Myostatin circulates in the blood and acts on muscle tissue to slow down further differentiation and growth of muscle cell precursors called myoblasts.
- In racehorses that were bred true, a mutation involving the substitution of a single nucleotide has been identified in the *MSTN* gene which codes for myostatin.
- At the site of this mutation, the DNA nucleotide has either a cytosine (C) base or a thymine (T) base, giving race horses three possible genotypes for this mutation: CC, CT or TT.
- At two years of age, racehorses with the *MSTN* CC genotype have greater muscle mass than those with the TT genotype.

Which of the following statements may be best concluded from the above?

- A** Racehorses with the CC genotype contain hyperactive myostatin protein in their muscle tissue.
- B** In racehorses with the CC genotype, there is a change in the reading frame of the mRNA codon from the site of the mutation.
- C** All racehorses with the CC genotype are able to run faster than those with the CT and TT genotype.
- D** Racehorses with the TT genotype develop more muscles later in life.

QUESTION 16

The *ras* oncogene is involved in a variety of human cancers. A student extracted DNA isolated from a number of normal and cancerous human tissues and digested the DNA with *EcoRI* restriction enzyme. A Southern blot was performed and the membrane was probed with radioactive-labelled *ras* DNA. After exposure to X-ray film, the resulting autoradiograph is shown below.



Which of the following could account for the results obtained?

- A *Ras* gene in normal tissue was mutated.
- B There is no *ras* mutation in cancer tissue 3.
- C *Ras* mRNA concentration is doubled in cancer tissue 1.
- D *Ras* DNA is amplified in cancer tissue 2.

QUESTION 17

CDC2 protein kinase is essential for mitosis to take place.

Which processes will help to maintain the presence of this protein?

- 1 methylation of guanine on the 5' end of mRNA
- 2 addition of adenine polynucleotides to the 3' end of mRNA
- 3 inhibition of cytoplasmic protease enzymes
- 4 inhibition of spliceosome involved in the removal of pre-mRNA introns

- A 1, 2 and 3 only B 2, 3 and 4 only C 1 and 3 only D 2 and 4 only

QUESTION 18

Which of the following statement(s) is/are true regarding hematopoietic stem cells and cancer cells?

- 1 Both are able to move from one location to another.
- 2 Both are found in cancer patients.
- 3 Both are specialised cells and capable to differentiate further.
- 4 Both are capable of indefinite replication.

A 2 only **B** 1 and 3 **C** 2 and 4 **D** 1, 2 and 4

QUESTION 19

The following statements describe Human Immunodeficiency Virus (HIV).

- 1 HIV genome undergoes high rate of mutation because the reverse transcriptase is error prone and has no proof reading.
- 2 HIV genome contains uracil, not thymine.
- 3 Gp120 enables HIV to enter the host cell by fusion of membranes.
- 4 Budding of the newly synthesized virions kills the host cell.

How many of these statement(s) about HIV is/are **incorrect**?

A 1 **B** 2 **C** 3 **D** 4

QUESTION 20

A type of bacteria causes fatalities in humans. These strains of bacteria possess genes for a toxin not found in the other harmless strains.

In an attempt to find out how these genes can be transferred between bacteria, several experimental set-ups were carried out. The results are shown in the table below.

Conditions	Results
DNA isolated from a virulent strain and incubated with another harmless strain.	Some virulent strains observed.
Virulent and harmless strains of bacteria incubated in a container with no barrier.	Some virulent strains observed.
Virulent and harmless strains of bacteria incubated in a container but separated by physical membrane barrier.	No virulent strain found in the side with harmless bacteria.

From the information provided only, which of the following gene transfer processes could have taken place?

- 1 Transformation
- 2 Conjugation
- 3 Transduction

A 1 only **B** 2 only **C** 1 and 2 only **D** 1, 2 and 3

QUESTION 21

The inheritance of flower colour in sweet pea plants is controlled by the interaction between two different genes. The biochemical pathway below illustrates the synthesis of purple pigments in these sweet pea plants.



Which of the following statements are **false**?

- 1 Recessive alleles at either gene loci in the homozygous recessive condition produce the same phenotype.
- 2 Inheritance of flower colour in sweet pea is an example of dominant epistasis.
- 3 Self-pollination of a double heterozygous sweet pea plant results in three different phenotypes.
- 4 Self-pollination of a double heterozygous sweet pea plant results in an offspring phenotypic ratio of 15:1.

- A 1 and 3 only
B 2 and 4 only
C 2, 3 and 4 only
D 1, 2 and 4 only

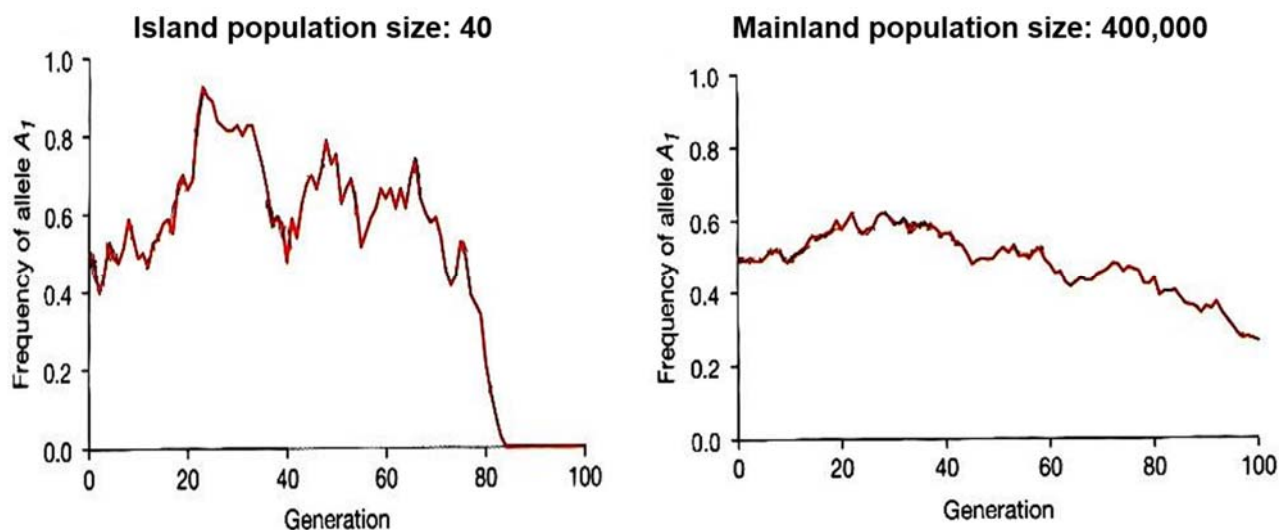
QUESTION 22

Which of the following correctly describes the function of second messengers in signal transduction pathway?

- A They relay the signal from the outside to the inside of the cell.
B They serve as transcription factors to activate the transcription process.
C They amplify the signal by phosphorylating other proteins.
D They bind to and alter the conformation of other effector proteins.

QUESTION 23

The graphs show the frequency of allele A_1 of two fruit fly populations on an island and on a nearby mainland over time.



Based on the information provided, which of the statement(s) is/are true?

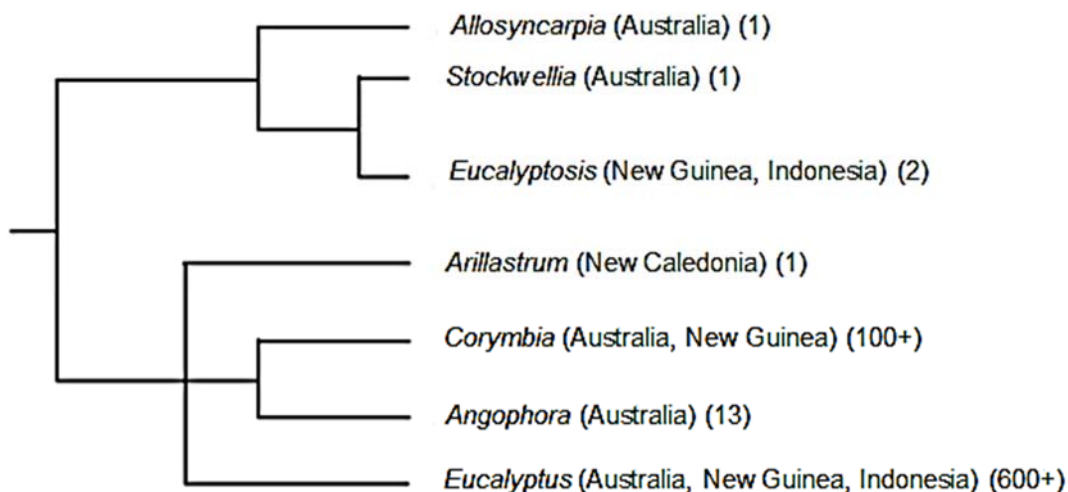
- 1 Selection pressures for gene A in the mainland changed over time.
- 2 Genetic drift and natural selection contributed to the change in frequency of allele A_1 in the mainland population.
- 3 The loss of allele A_1 from the island population could be due to a spontaneous mutation in the allele sequence.
- 4 Random chance could result in the fixation or loss of allele A_1 in the island population.

A 3 only **B** 3 and 4 only **C** 1, 2 and 4 **D** 1, 2, 3 and 4

QUESTION 24

Eucalyptus, commonly known as gum trees, are Australian trees. They have been recently classified into seven different genera on the basis of nuclear and chloroplast DNA sequencing.

A proposed phylogeny for the seven genera is shown in the diagram, along with the countries in which they are found.



key: Numbers in brackets () refer to the numbers of species in the genus.

Which of the following statements is correct?

- A *Eucalyptosis* evolved from *Allosyncarpia* as they share a more recent common ancestor.
- B Speciation in *Eucalyptus* was assisted by different selecting pressures.
- C The similarities between the members of each species are less than the similarities between the members of each genus.
- D The genus that evolved most recently was *Angophora*.

QUESTION 25

The skin of a small amphibian contains a substance, tetrodotoxin (TTX), that is deadly to all predators except the common garter snake. TTX acts by blocking membrane ion channels involved in muscular movement. Some of the garter snakes slow down for a few hours after eating the amphibian, but are not otherwise harmed.

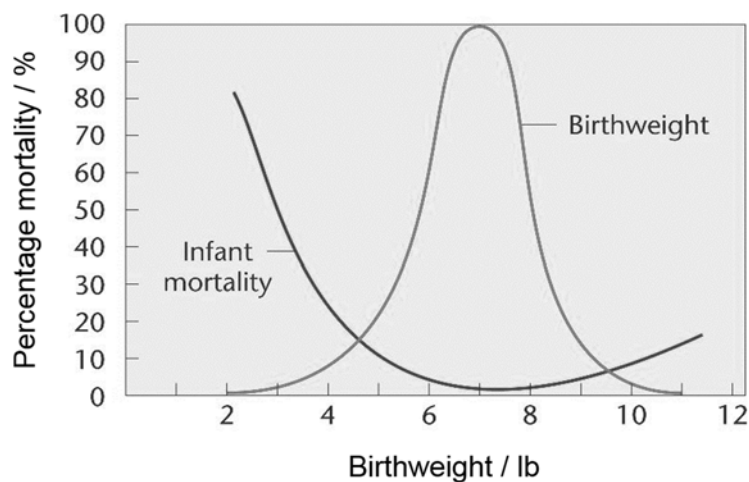
Some variants of the amphibian have higher toxicity because of higher TTX production, while some variants of the garter snake have higher resistance. Where snakes have not been exposed to TTX, the snake variants with higher resistance have been observed to move more slowly than those with normal resistance.

From the information provided above, what is **not** a valid suggestion concerning natural selection and evolution in the amphibian and the garter snake?

- A A selection pressure acting against the garter snake with higher resistance is the amphibian variant with normal toxicity.
- B Location with the most toxic amphibian variants are likely to have the most resistant garter snake variants.
- C Predation by garter snake is the selection pressure for increasing toxicity of the pray; the selection pressure acting against the most resistant garter snake is predation.
- D TTX resistance is likely to involve a mutation in the gene coding for the membrane ion channels.

QUESTION 26

The graph below shows the relationship between birthweight and infant mortality in humans.



What type of selection is demonstrated above?

- A Directional selection
- B Disruptive selection
- C Stabilising selection
- D Artificial selection

QUESTION 27

Bacteria that infect skin wounds can become resistant to the antibiotic commonly used to kill them.

Which statement explains how these bacteria could become resistant to the antibiotic?

- A Exposure to the antibiotic causes a change in the bacterial cell wall, preventing the antibiotic from working.
- B Other bacteria that normally protect the skin against infection are killed by the antibiotic, allowing the infectious bacteria to survive.
- C Random mutations in DNA allow some of the bacteria to survive in the presence of the antibiotic.
- D The antibiotic causes the bacterial DNA to mutate, allowing the bacteria to survive in the presence of the antibiotic.

QUESTION 28

A vaccine is used to create artificial active immunity. After being given a vaccine, it will take a period of time before a person develops long-term immunity against the disease.

Which statement about this period of time explains this delay?

- A No memory cells have been produced from B-lymphocytes.
- B No plasma cells have been produced from B-lymphocytes.
- C The primary immune response has not produced enough antibodies.
- D The secondary immune response has not produced enough antibodies.

QUESTION 29

Levels of carbon dioxide in the atmosphere fall during summer in the northern hemisphere.

What best explains this trend?

- A seasonal decrease in the use of fossil fuel and wood for heating
- B seasonal increase in the amount of carbon dioxide dissolved in the oceans
- C seasonal increase in the rate of decay of organic matter
- D seasonal increase in the rate of photosynthesis in the northern hemisphere

QUESTION 30

By which mechanism do greenhouse gases contribute to global warming?

- A Their higher concentration absorbs more long wave radiation coming from the sun.
- B Short wave radiation emitted from the Earth's surface increases with their concentration.
- C They absorb higher amounts of long wave radiation emitted from the Earth's surface as their concentration increases.
- D They absorb higher amounts of short waves generated by an increase in combustion of fossilized organic matter.

End of Paper

H2 Biology Prelim Paper 1 Answers

1. C	6. C	11. C	16. D	21. C	26. C
2. D	7. C	12. B	17. A	22. D	27. C
3. A	8. C	13. A	18. C	23. C	28. A
4. D	9. C	14. B	19. B	24. B	29. D
5. B	10. C	15. D	20. C	25. A	30. C